

A Systematic Explainability Framework for Hybrid Spatio-Temporal Models based on Integrated Gradients^{*}

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Abstract. Hybrid deep learning architectures combining convolutional and recurrent components are increasingly adopted for modelling complex spatio-temporal processes. Despite their strong predictive performance, these models typically operate as black boxes, hindering their deployment in safety-critical and decision-support systems. This paper proposes a general explainability framework for hybrid temporal architectures based on Integrated Gradients, enabling systematic attribution of model predictions to both input features and historical time steps. Unlike architecture-specific interpretability mechanisms, the proposed approach is model-agnostic and can be directly applied to a wide range of convolutional–recurrent networks without requiring retraining or structural modifications.

As a proof of concept, we apply the framework to a previously validated CNN–BiLSTM model for long-term urban traffic forecasting and perform a multi-horizon, multi-location explainability analysis. The resulting attributions illustrate how contextual variables and temporal dependencies are leveraged by the model across prediction horizons. While the empirical evaluation focuses on traffic data, the proposed methodology is broadly applicable to other hybrid deep learning models operating on multivariate time series. Overall, this work provides a practical and theoretically grounded approach for enhancing transparency in hybrid architectures, facilitating their adoption in real-world predictive systems.

Keywords: Explainable Artificial Intelligence, Hybrid Deep Learning, Integrated Gradients, Traffic Flow Forecasting

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1 Introduction

The paradigm of hybrid deep learning, which integrates functionally diverse architectures to process complex data structures, has established a new state-of-the-art in predictive modelling across multiple spatio-temporal domains [24]. By leveraging the complementary strengths of heterogeneous components, these systems excel in capturing the non-linear dynamics inherent in high-dimensional datasets [23]. However, as these architectures grow in complexity, their *black-box* nature becomes a significant barrier to their deployment in critical environments where accountability and transparency are as essential as predictive precision [7].

Within the field of Intelligent Transportation Systems, this lack of interpretability is particularly problematic. Traffic flow forecasting is a fundamental component for decision-making in infrastructure planning and urban mobility policies. While deep learning has led to substantial improvements in short-term accuracy, long-term forecasting remains a challenging task due to the influence of contextual and non-recurrent factors, such as weather conditions and roadway interactions. In this context, the field of Explainable Artificial Intelligence (XAI) has emerged as a crucial discipline to address the inherent opaqueness of these predictive models [3].

Despite the proliferation of interpretability methods, providing meaningful insights into multi-component structures remains an open challenge. Traditional techniques, such as LIME or SHAP, were primarily designed for static or tabular data and often struggle to account for the intricate interplay between spatial hierarchies and temporal evolutions [2]. This gap is particularly evident in long-term forecasting, where feature relevance is not static but fluctuates across multiple temporal resolutions.

In our previous work [15], we addressed the accuracy challenge by introducing a CNN-BiLSTM hybrid model optimized for long-term urban forecasting. While this architecture significantly outperformed classical baselines, its internal reasoning remained opaque. This limitation is not merely academic; public authorities require transparent models to justify automated interventions and build systemic trust [14].

To bridge this gap, this paper moves beyond predictive performance to introduce a systematic explainability framework based on Integrated Gradients (IG) [21]. IG is a gradient-based attribution technique that satisfies the axiom of completeness, attributing model predictions to input features by integrating gradients along a continuous path from a baseline to the actual observation.

Rather than proposing a new predictive architecture, our objective is to gain *white-box* insights into the internal logic of the already validated CNN-BiLSTM model. Using the same experimental setup and prediction horizons (12 to 72 hours) as in our earlier study, we analyse how the contributions of traffic-related variables, contextual information, and historical observations evolve. This allows us to scrutinize how feature relevance and spatio-temporal dependencies shift across different urban scenarios, moving past the high-level, aggregate explanations provided by existing studies.

In conclusion, the primary contribution of this work is a validated methodology for interpreting complex hybrid models. Although grounded in urban traffic flow, the proposed framework is designed to be model-agnostic across the class of CNN-RNN hybrids, effectively bridging the gap between state-of-the-art deep learning and trustworthy AI.

The main contributions of this paper are the following:

- We introduce a general explainability framework for hybrid deep learning models based on Integrated Gradients.
- We analyse how the relevance of input variables and historical time steps evolves across different prediction horizons, specifically for long-term settings.
- We provide an in-depth analysis of the convolutional block’s role, demonstrating that it learns meaningful latent representations of the urban traffic context.
- We demonstrate that the proposed analysis enhances the interpretability and practical utility of deep learning models for real-world traffic management applications.

The code, data and experimental results used in this work are available at https://github.com/MMH1997/CNN-BiLSTM-network/tree/main/XAI_IG.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents Integrated Gradients, the methodology employed in this study. Section 4 briefly describes the model and datasets used. Section 5 reports the experimental setup and discusses the obtained results. Finally, Section 6 presents the conclusions and outlines directions for future work.

2 Related Work

Deep learning techniques have been extensively applied to urban traffic forecasting due to their capability to model complex non-linear temporal dynamics. Early approaches primarily relied on recurrent neural networks, particularly LSTM-based architectures, to capture sequential dependencies in traffic flow data [22,25]. Subsequent works incorporated convolutional layers to extract local spatial or feature-level patterns, leading to hybrid CNN–LSTM models that demonstrated improved performance in multivariate traffic prediction tasks [5].

More recently, bidirectional recurrent architectures have been explored to enhance temporal representation by exploiting information from both past and future contexts [11,1]. In our previous work [15], we proposed a CNN–BiLSTM architecture for urban traffic prediction across multiple streets and temporal resolutions, achieving competitive predictive accuracy. The present study builds directly upon this model, extending it with explainability capabilities.

In parallel, explainable artificial intelligence has gained increasing attention in time-series analysis and intelligent transportation systems. Several studies incorporate attention-based mechanisms within traffic prediction models to

highlight relevant temporal segments or input variables [26,20]. While these approaches provide internal interpretability signals, they require architectural modifications and do not offer formal guarantees regarding attribution faithfulness.

Other works adopt post-hoc explainability techniques, such as SHAP to identify influential variables in traffic forecasting [16,10]. Although these methods provide useful insights, they often rely on heuristic perturbations and lack theoretical consistency properties when applied to deep temporal models.

Integrated Gradients (IG) [21] offers an axiomatic framework for feature attribution by integrating gradients along a continuous path between a baseline input and the observed sample. The method satisfies desirable theoretical properties such as sensitivity and completeness. While IG has been occasionally applied to convolutional neural networks in computer vision [17,18], its use in recurrent architectures for time-series analysis remains limited [19]. Moreover, existing applications typically focus on either CNN or LSTM models in isolation.

While gradient-based attribution has been previously applied to time series [8], its use in complex urban forecasting remains limited. Most existing literature in intelligent transportation systems either focuses on intrinsic interpretability of simpler models or provides static explanations for short-term horizons [6], which may not fully capture the temporal dynamics of long-term forecasting. Our work distinguishes itself by providing a systematic application of IG specifically tailored to disentangle the internal reasoning of CNN-BiLSTM models across expanding temporal horizons (up to 72 hours). To the best of our knowledge, this constitutes the first study to map these attribution patterns across multiple urban scenarios and temporal resolutions simultaneously, providing a more granular understanding of the model’s predictive logic.

3 Integrated Gradients Methodology

In this section we present the main features of our IG methodology. IG [21] is a gradient-based attribution method designed to explain the prediction of differentiable models by assigning relevance scores to each input feature. The method quantifies how much each input contributes to a specific prediction by integrating gradients along a continuous path between a reference input (baseline) and the actual input.

Let $x \in \mathbb{R}^n$ denote the input vector and $F(x)$ the corresponding model output. Let x' denote a baseline input vector representing a neutral or reference state of the model. The attribution of the i -th feature is defined as:

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

where $\alpha \in [0, 1]$ is a scalar interpolation parameter. In our experiment, this interval is discretized into 50 equally spaced points to numerically approximate the path integral.

The previous formulation accumulates the gradients of the model output with respect to each input feature as the input transitions from the baseline

to the observed sample. Rather than computing gradients at a single point, IG integrates gradients along the entire path, providing more stable and reliable attributions.

IG satisfies several desirable axiomatic properties, including sensitivity and completeness [21]. Sensitivity ensures that any input feature whose variation changes the model prediction receives a non-zero attribution, preventing influential variables from being ignored. The completeness property ensures that the sum of all attribution values approximates the difference between the model prediction and the baseline output:

$$\sum_i IG_i(x) \simeq F(x) - F(x')$$

In practice, the continuous integral is approximated using a standard Riemann sum, that is, by averaging gradient values computed at a finite number of uniformly spaced interpolation points along the interval $[0, 1]$. In this work, 50 interpolation points are employed. A zero-valued tensor is used as baseline. Since all input variables are normalized, this corresponds to the empirical mean of each feature. For each input sample, intermediate points are generated along a straight-line interpolation between the baseline and the actual input. Gradients are computed for all interpolated samples simultaneously and averaged to obtain the final attribution scores.

Absolute attribution values are retained to focus on the magnitude of feature influence independently of direction. IG is computed over 100 test samples to obtain statistically robust explanations while maintaining computational feasibility.

Algorithm 1 summarizes the IG framework that we present in this work.

Algorithm 1 Integrated Gradients

Require: Model F , input sample x , baseline x' (default zero)

Ensure: Attribution scores $IG(x)$

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1:  $m \leftarrow 50$ 
2: if  $x'$  is not provided then
3:    $x' \leftarrow 0$ 
4: end if
5: for  $i = 0$  to  $m$  do
6:    $\alpha \leftarrow i/m$ 
7:    $x^{(\alpha)} \leftarrow x' + \alpha(x - x')$ 
8: end for
9: Compute gradients  $\nabla F(x^{(\alpha)})$  for all interpolated samples simultaneously
10: AvgGrads  $\leftarrow$  average gradients over all steps
11:  $IG(x) \leftarrow (x - x') \odot$  AvgGrads
12: Output  $IG(x)$ 
    
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4 Original model and dataset

In this section, we will briefly describe the datasets employed in this work and the CNN-BiLSTM model in which the explainability analysis is performed.

4.1 Datasets

In our previous work [15], we used 16 different datasets for our experiments. Each of them corresponds to a specific combination of one street of Madrid and one temporality, obtained from four target streets of Madrid and four different temporalities (12, 24, 48 and 72 hours). For each dataset, input samples are constructed using four historical observations separated by a temporal resolution Δ . Specifically, the target corresponds to the traffic flow at time t , while the input window is formed by samples at $t - \Delta$, $t - (\Delta + 1)$, $t - (\Delta + 2)$, and $t - (\Delta + 3)$. This formulation enables the model to capture periodic temporal dependencies associated with different temporal horizons.

At each hourly time step, nine explanatory variables are considered, resulting in a total of 36 input features per sample. These variables include the traffic flow at the target street, traffic flow at two nearby auxiliary streets, four meteorological variables (daily mean temperature, daily rainfall, and daily minimum and maximum temperatures), a day-type indicator, and the hour of day. All variables are normalized using min-max scaling based on the statistics of the training set, following the preprocessing strategy adopted in our previous work. The CNN-BiLSTM model was trained beforehand and IG is subsequently applied as a post-hoc explainability method on the trained model. Finally, the target variable is defined as the traffic flow at the target street at time t .

The datasets that we used in this work are publicly available at <https://github.com/MMH1997/CNN-BiLSTM-network>. All these dataset include data from January 2018 to July 2021. We do not use more current data because, as explained before, we do not want to modify the hybrid model that we constructed in our previous work.

4.2 CNN-BiLSTM model

The predictive model used in this work corresponds to the CNN-BiLSTM architecture proposed in our previous study [15]. The model combines a convolutional neural network for feature extraction with a bidirectional LSTM for temporal modelling. The CNN block learns latent representations from the multivariate input, while the BiLSTM captures temporal dependencies by processing the extracted features in both forward and backward directions. A fully connected layer produces the final traffic flow prediction at the target street.

All architectural settings and hyperparameters are adopted from the original work without modification. In this study, the CNN-BiLSTM model is treated as a black-box predictor and IG is applied to analyse the contribution of each input variable and temporal lag to the model output. A schematic representation of the CNN-BiLSTM architecture is shown in Figure 1.

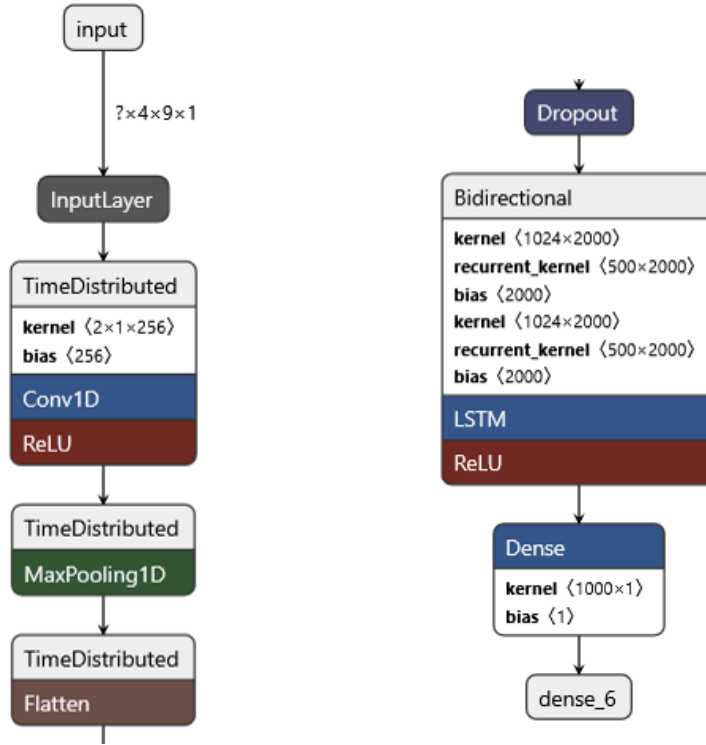


Fig. 1: CNN-BiLSTM model architecture (first and second part)

5 Experiments and Results

In this section, we present the experimental setup and discuss the explainability results obtained using IG. We analyse feature relevance and temporal attribution patterns across multiple streets and temporal resolutions in order to provide insights into the internal behaviour of the CNN-BiLSTM model.

5.1 Experiment Setup

The experiments are conducted using the same 16 datasets, model architecture, and preprocessing pipeline used in our previous work. Each dataset corresponds to a specific combination of urban street and temporal resolution. Input samples are constructed from four historical observations separated by a temporal resolution Δ , and consist of nine explanatory variables per time step, resulting in 36 input features per sample. The target variable is defined as the traffic flow at the target street at time t .

All variables are normalized using min-max scaling based on training set statistics. The CNN-BiLSTM models are trained beforehand following the orig-

inal experimental protocol, and IG is subsequently applied as a post-hoc explainability method on the pre-trained networks.

For each dataset, IG is computed over 100 randomly selected test samples to obtain robust attribution estimates while maintaining computational feasibility. A zero-valued tensor is used as baseline, corresponding to the empirical mean of each normalized feature. The path integral is approximated using 50 interpolation steps between the baseline and each input sample.

To enable comparative analysis across temporal resolutions and urban locations, attribution values are aggregated by averaging absolute attributions over samples and temporal lags. Since all experiments share the same architecture, preprocessing pipeline, and feature semantics, relative attribution patterns can be compared across datasets. However, absolute attribution magnitudes are not directly comparable between different streets due to inherent differences in local traffic dynamics. Therefore, our analysis focuses on relative importance trends and attribution rankings within each dataset.

5.2 Results

In this section, we describe the results obtained after applying IG over the previously described datasets.

First, we analyse the global contribution of all input variables by aggregating XAI attributions across streets, horizons, and time-steps. As we stated previously, since raw attribution values are not directly comparable across models, each dataset was normalized such that the total attribution mass equals one, resulting in relative importance scores.

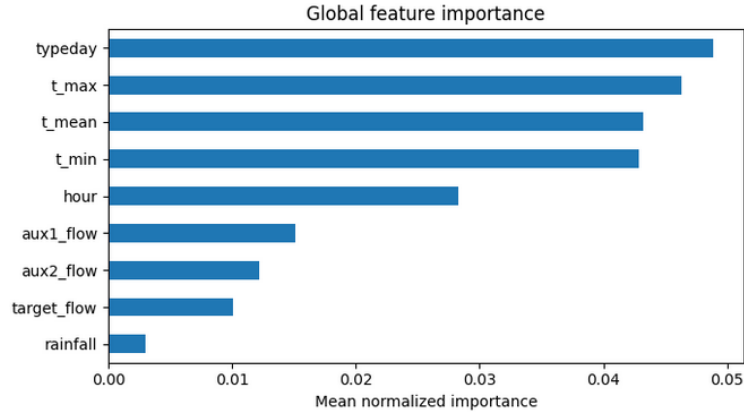


Fig. 2: Global average importance of each variable.

Figure 2 shows the average importance of each feature over all experimental configurations. The type of day emerges as the most influential feature followed

by temperature-related variables. Hour also exhibits a substantial relevance. In contrast, rainfall contributes minimally. These results indicate that the model primarily relies on temporal and thermal context rather than direct traffic history, suggesting a strong dependency on periodic and meteorological patterns.

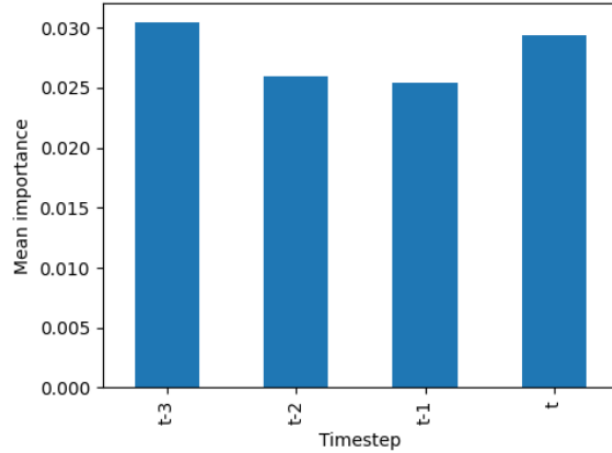


Fig. 3: Average importance of each horizon.

According to time-step, as can be seen at Figure 3, feature attributions exhibit a non-monotonic temporal pattern. The current time-step (t) and the earliest lag ($t-3$) present comparable and dominant importance (on average, 0.0294 and 0.0304 respectively), whereas intermediate time-steps ($t-1$ and $t-2$) show consistently lower contribution (on average, 0.0254 and 0.0259 respectively). This distribution suggests that the model simultaneously exploits immediate information and longer-range historical context, while short-term intermediate states provide limited additional explanatory value. Such behaviour may indicate delayed or periodic dependencies in traffic dynamics, potentially linked to daily mobility cycles or redundancy among consecutive observations.

Figure 4 shows the mean normalized importance of each variable across streets. A highly consistent global hierarchy is observed, with type of day and temperature-related variables dominating in all locations, followed by hour, while traffic-related variables exhibit comparatively low relevance and rainfall remains negligible. Although minor quantitative differences appear across streets, reflecting local sensitivities, the overall explanatory structure is remarkably stable. This indicates that the model relies on a common set of driving factors across locations, with street-specific effects acting as secondary modifiers.

Similarly to the street-level analysis, a consistent global hierarchy of feature importance can be observed across all prediction horizons at Figure 5. Although relative contributions vary slightly with horizon, the overall explanatory struc-

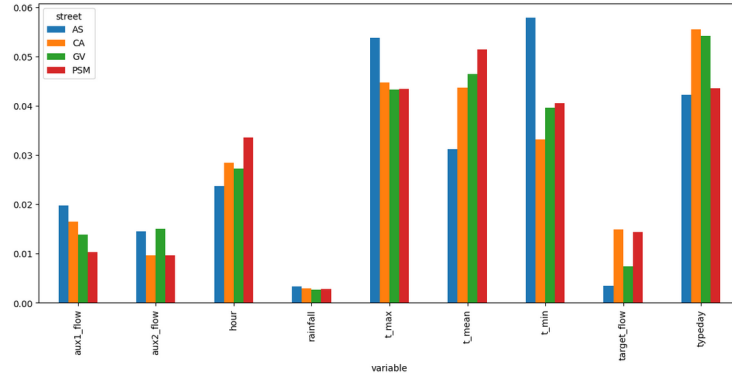


Fig. 4: Average importance of each variable by street.

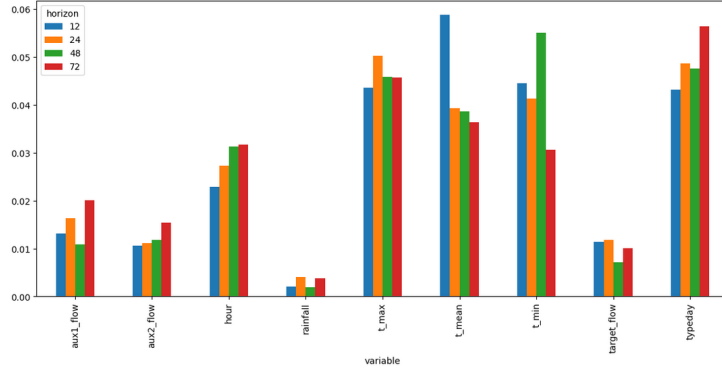


Fig. 5: Average importance of each variable by street.

ture is preserved, indicating that horizon primarily modulates feature weights rather than altering the underlying prediction mechanism.

To extend our result analysis, we divide the input variables into three groups: weather-related, calendar-related, and traffic flow-related. Figure 6 summarizes the average normalized importance of each group across streets. While weather and flow-related features remain relatively stable across streets, calendar variables exhibit greater variability, indicating stronger street-specific sensitivity to temporal patterns. Nevertheless, weather features dominate in all cases, followed by calendar inputs, whereas traffic flows remain marginal, suggesting that spatial variability is mainly driven by calendar effects.

Finally, we present a two-dimensional PCA projection of the normalized attribution vectors at Figure 7. clear organization primarily driven by prediction horizon emerges, whereas datasets from different streets largely overlap, confirming the presence of a shared explanatory structure across locations. Longer horizons tend to occupy more extreme regions of the latent space, indicating

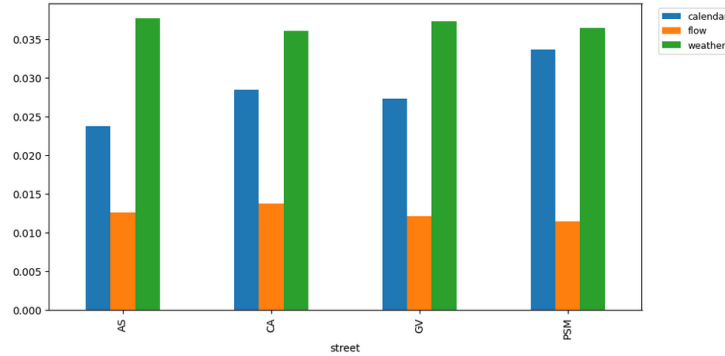


Fig. 6: Average feature importance by variable type across streets

increasing reliance on contextual features, while spatial differences appear secondary. AS72 (Arturo Soria Street, 72 hours horizon) appears isolated in the latent space, reflecting a distinct attribution profile at longer horizons. Overall, these results suggest that forecast horizon constitutes the dominant factor shaping explanation patterns, with street-specific effects introducing only minor variations.

Overall, the explainability analysis reveals a globally consistent attribution structure dominated by calendar and temperature-related features, with spatial variability playing a secondary role and prediction horizon emerging as the primary driver of explanatory differences, indicating that the forecasting process is governed more by recurring temporal and environmental patterns than by direct traffic autoregression.

5.3 Analysis of experimental results

The experimental results show that calendar-related and weather-related variables dominate the prediction process across all streets and horizons, while direct traffic-flow inputs play a clearly secondary role.

As shown in Figure 2, the type of day variable presents the highest average normalized attribution ($\simeq 0.045$), exceeding rainfall ($\simeq 0.01$) by more than a factor of four and doubling the contribution of individual traffic flow variables ($\simeq 0.015 - 0.02$). Temperature-related features also exhibit consistently high relevance ($\simeq 0.035 - 0.04$ each), confirming the strong influence of environmental context.

Temporal attribution further reveals a non-monotonic pattern: the current time step (t) and the most distant lag ($t - 3$) exhibits the largest contributions ($\simeq 0.0294$ and 0.0304 respectively), while intermediate lags ($t - 1$ and $t - 2$) are less influential ($\simeq 0.0254 - 0.0259$). This behaviour indicates that the model simultaneously leverages immediate information and longer-range temporal dependencies, which is consistent with periodic mobility dynamics. Importantly, this observation supports the suitability of the CNN-BiLSTM architecture, as

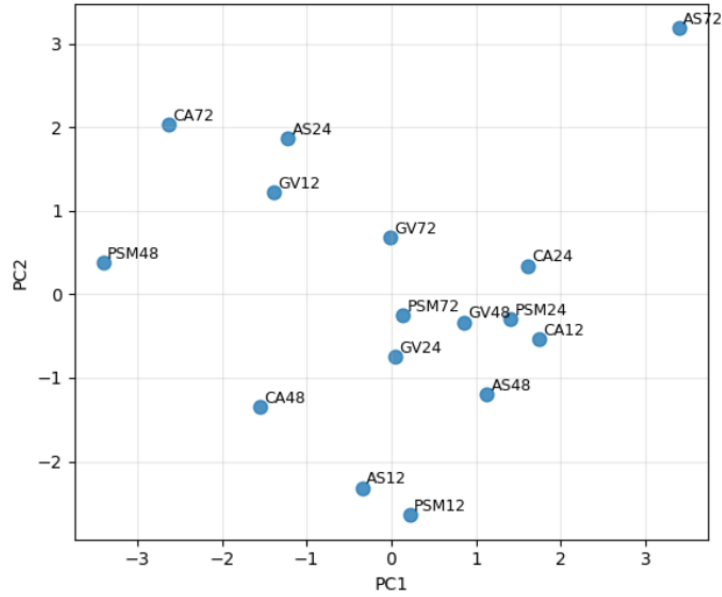


Fig. 7: PCA projection of normalized attribution vectors

the convolutional block captures local feature interactions while the bidirectional LSTM effectively models non-adjacent temporal relationships, enabling the joint exploitation of short-term cues and longer-term contextual patterns.

Figure 4 shows that importance remains highly consistent across streets suggesting a shared explanatory structure with only minor location-specific variations. Similarly, Figure 5 and the PCA projection in Figure 7 show that prediction horizon is the main driver of attribution variability, while spatial differences are secondary.

Finally, as can be seen in Figure 6, grouping features by type shows that weather-related inputs account for approximately 35 – 38% of the total attribution mass, followed by calendar variables (25 – 30%) whereas traffic-flow features contribute only 10 – 15%. Overall, these results indicate that long-term forecasting in this setting is governed primarily by recurring temporal and environmental patterns rather than direct traffic autoregression, validating the effectiveness of the proposed explainability framework in revealing the internal logic of hybrid CNN-BiLSTM models.

6 Conclusions and future work

This paper presented an explainability framework based on IG for hybrid deep learning architectures combining convolutional and recurrent components. The main contribution of this work is a model-agnostic post-hoc attribution methodology that enables the joint analysis of feature relevance and temporal dependen-

cies in complex spatio-temporal models, without requiring architectural modifications or additional training procedures. Although the experimental evaluation focuses on a CNN-BiLSTM architecture for long-term urban traffic forecasting, the proposed framework is not specific to this model or application domain. Instead, it can be directly applied to other hybrid architectures operating on multivariate time series, offering a unified mechanism for explainability in settings where interpretability remains a challenge.

The experimental results demonstrate the practical feasibility of the approach and reveal consistent attribution patterns across streets and prediction horizons. Calendar and temperature-related variables emerge as the dominant explanatory factors, while prediction horizon is identified as the main driver of variability in attribution profiles. These findings provide insights into how contextual information and historical observations are leveraged by the model and highlight the value of the proposed framework for understanding hybrid architectures in long-term forecasting scenarios.

The explainability code and experimental results are available in a dedicated subfolder of the CNN-BiLSTM repository used in our previous work: https://github.com/MMH1997/CNN-BiLSTM-network/tree/main/XAI_IG.

Future work will extend the proposed framework to alternative hybrid architectures and application domains beyond traffic forecasting, including energy demand [4], air quality prediction [12,13], and environmental time-series analysis [9]. We also plan to evaluate complementary explainability techniques and baseline strategies to assess attribution robustness. Finally, integrating explanation outputs into operational decision-support systems constitutes a promising direction for translating model insights into practical guidance for practitioners and policy makers.

References

1. Abduljabbar, R.L., Dia, H., Tsai, P.W.: Unidirectional and bidirectional LSTM models for short-term traffic prediction. *Journal of Advanced Transportation* **2021**(1), 5589075 (2021)
2. Abusitta, A., Li, M.Q., Fung, B.C.: Survey on Explainable AI: Techniques, challenges and open issues. *Expert Systems with Applications* **255**, 124710 (2024). <https://doi.org/10.1016/j.eswa.2024.124710>
3. Adadi, A., Berrada, M.: Peeking inside the black-box: A survey on Explainable Artificial Intelligence (XAI). *IEEE Access* **6**, 52138–52160 (2018). <https://doi.org/10.1109/ACCESS.2018.2870052>
4. Aslam, S., Herodotou, H., Mohsin, S.M., Javaid, N., Ashraf, N., Aslam, S.: A survey on deep learning methods for power load and renewable energy forecasting in smart microgrids. *Renewable and Sustainable Energy Reviews* **144**, 110992 (2021)
5. Bogaerts, T., Masegosa, A.D., Angarita-Zapata, J.S., Onieva, E., Hellinckx, P.: A graph CNN-LSTM neural network for short and long-term traffic forecasting based on trajectory data. *Transportation Research Part C: Emerging Technologies* **112**, 62–77 (2020)
6. Gaur, L., Sahoo, B.M.: *Explainable Artificial Intelligence for Intelligent Transportation Systems*. Springer (2022). <https://doi.org/10.1007/978-3-031-09644-0>

7. Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D.: A survey of methods for explaining black box models. *ACM Computing Surveys* **51**(5), 1–42 (2018). <https://doi.org/10.1145/3236009>
8. Ismail, A.A., Gunady, M., Bravo, H.C., Feizi, S.: Benchmarking deep learning interpretability in time series predictions. In: 34th International Conference on Neural Information Processing System. pp. 6441–6452 (2020). <https://doi.org/10.5555/3495724.3496264>
9. Kaur, J., Parmar, K.S., Singh, S.: Autoregressive models in environmental forecasting time series: a theoretical and application review. *Environmental Science and Pollution Research* **30**(8), 19617–19641 (2023)
10. Khader, S., Awajan, A.: Interpretable traffic flow prediction using SHAP and STID-Recurrence Model. In: 2025 International Conference on New Trends in Computing Sciences (ICTCS). pp. 211–218. IEEE (2025)
11. Ma, C., Dai, G., Zhou, J.: Short-term traffic flow prediction for urban road sections based on time series analysis and LSTM-BiLSTM method. *IEEE Transactions on Intelligent Transportation Systems* **23**(6), 5615–5624 (2021)
12. Méndez, M., Merayo, M.G., Núñez, M.: Machine learning algorithms to forecast air quality: a survey. *Artificial Intelligence Review* **56**(9), 10031–10066 (2023). <https://doi.org/10.1007/s10462-023-10424-4>
13. Méndez, M., Montero, C., Núñez, M.: Using deep transformer based models to predict ozone levels. In: 14th Asian Conference on Intelligent Information and Database Systems, ACIIDS’22, LNAI 13757. pp. 169–182. Springer (2022). https://doi.org/10.1007/978-3-031-21743-2_14, https://doi.org/10.1007/978-3-031-21743-2_14
14. Mersha, M., Lam, K., Wood, J., AlShami, A.K., Kalita, J.: Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction. *Neurocomputing* **599**, 128111 (2024). <https://doi.org/https://doi.org/10.1016/j.neucom.2024.128111>
15. Méndez, M., Merayo, M.G., Núñez, M.: Long-term traffic flow forecasting using a hybrid CNN-BiLSTM model. *Engineering Applications of Artificial Intelligence* **121**, 106041 (2023)
16. Punia, D., Sharma, A.K., et al.: Next-generation computing for smarter traffic: An Explainable AI approach to short-term flow prediction. In: 2025 IEEE International Conference on Computer, Electronics, Electrical Engineering & their Applications (IC2E3). pp. 1–6. IEEE (2025)
17. Qi, Z., Khorram, S., Li, F.: Visualizing deep networks by optimizing with integrated gradients. In: 34th AAAI conference on artificial intelligence. pp. 11890–11898 (2020)
18. Rabbah, J., Ridouani, M., Hassouni, L.: Improving pneumonia diagnosis with high-accuracy cnn-based chest x-ray image classification and integrated gradient. *Biomedical Signal Processing and Control* **101**, 107239 (2025)
19. Santoso, N., Mendonça, I., Aritsugi, M.: Text augmentation based on integrated gradients attribute score for aspect-based sentiment analysis. In: 2023 IEEE International Conference on Big Data and Smart Computing (BigComp). pp. 227–234. IEEE (2023)
20. Shi, X., Qi, H., Shen, Y., Wu, G., Yin, B.: A spatial-temporal attention approach for traffic prediction. *IEEE Transactions on Intelligent Transportation Systems* **22**(8), 4909–4918 (2020)
21. Sundararajan, M., Taly, A., Yan, Q.: Axiomatic attribution for deep networks. In: International conference on machine learning. pp. 3319–3328. PMLR (2017)

22. Tian, Y., Zhang, K., Li, J., Lin, X., Yang, B.: LSTM-based traffic flow prediction with missing data. *Neurocomputing* **318**, 297–305 (2018)
23. Wang, S., Cao, J., Yu, P.S.: Deep learning for spatio-temporal data mining: A survey. *IEEE Transactions on Knowledge and Data Engineering* **34**(8), 3681–3700 (2022). <https://doi.org/10.1109/TKDE.2020.3025580>
24. Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., Yu, P.S.: A comprehensive survey on graph neural networks. *IEEE Transactions on Neural Networks and Learning Systems* **32**(1), 4–24 (2021). <https://doi.org/10.1109/TNNLS.2020.2978386>
25. Yang, B., Sun, S., Li, J., Lin, X., Tian, Y.: Traffic flow prediction using LSTM with feature enhancement. *Neurocomputing* **332**, 320–327 (2019)
26. Zheng, C., Fan, X., Wang, C., Qi, J.: Gman: A graph multi-attention network for traffic prediction. In: 34th AAAI conference on artificial intelligence. pp. 1234–1241 (2020)